



IIC AGRICULTURE STARTUP

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KRISHIGYAN

Smart Hyper-Local Crop Advisory for Small & Marginal Farmers

"SUPPORT FARMERS AT EVERY STAGE AND INSURE THEIR FUTURE RESILIENCE"

SOFTWARE AS A SERVICE

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Problem

10 – 15 % yield lost every year due to lack of right information

Weather and government advisories are often long and technical.

Critical warnings may arrive late or be ignored.

Weather, crop advice, schemes and compensation are spread across platforms.

What Unique in Our Solution

1. **Video-First** : 30–45 sec visual advisories instead of long, technical messages or PDFs.
2. **AI Tutor in the Farmer's Language** : A familiar AI avatar explains the advice through regional voice and simple language, making it easier to understand.
3. **Advice → Action → Support** : We don't stop at prediction—we tell farmers what to do today and guide them toward compensation and government schemes when things go wrong.



Key Value

Proposition "One Day, One Decision"

Instead of giving farmers 10 different advisories, Krishi Gyan converts all relevant data into one clear answer: "Aaj mere khet mein mujhe kya karna chahiye?"

Video Advisory using Avatar

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Solution

KrishiGyan turns complex weather, soil and crop data into simple, hyper-local video advice in the farmer's own language - so they know what to do, when to do it, and avoid costly mistakes.

Easy Access

Regional Language support

Hyper – local Insights

End-To-End Support

Farmers

Useful Information according to their area at right time

Better Crops

1. Data Sources



Weather Forecasts
(IMD APIs)
Rainfall , Temperature,
Humidity, Alerts



Soil Data
(ISRO Bhuvan , Soil Health
Card)
Soil type , Nutrients level



Crop Information
(ICAR , State Agri Dept.)
Crop Calendar ,
Best Practices



Government Schemes
(PMFBY , e-NAM , State
portals)
Eligibility and Guidelines



Farmer Location
(GPS/Village/Block)
Hyper-local personalization

2. Processing and AI engine



Data Integration
Collect , clean and merge
data from different sources



AI Analysis & Prediction
Weather risk , Crop
Suitability , Pest/Disease risk
, Yield insights



Script Generation
Create short, simple and
actional message(LLM)



Text-to-Speech
Convert to regional language
audio(multi-dialect)



Video Generation
AI avatar + Visuals+
Subtitles (30-45 seconds)

3. Personalized Outputs



AI Video Advisory
30-45 sec video in
farmers language



Risk Maps
Rainfall , flood,
drought and pest
alerts



Crop Guidance
Best crop for your
area and season



Scheme Assistance
Apply for the
compensation and
government schemes

4. Delivery to Farmers



WhatsApp
Emergency and recommendation
message



Lightweight Mobile APP
Easy to Download and Use



- ✓ Know the weather
- ✓ Take the right action
- ✓ Protect your crops

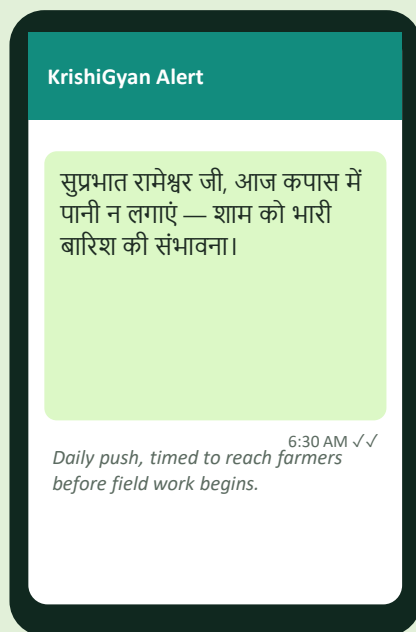
5. Feedback and Continuous Improvement

Farmers responses , crop outcomes and field data to make the system
smarter

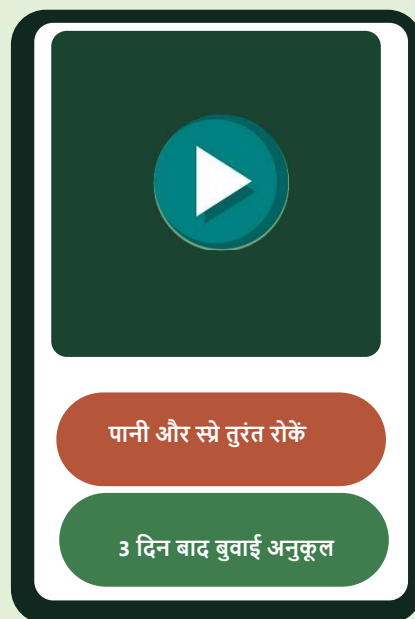
Implementation & User Experience

Designed for a low-end smartphone also

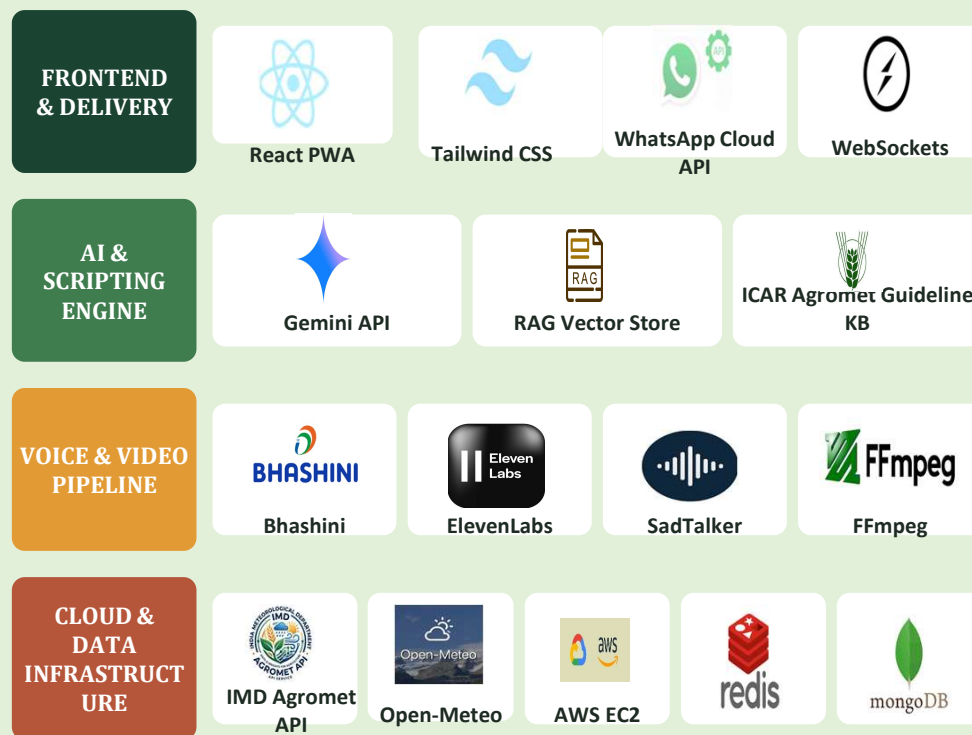
Screen 1 — Morning Alert



Screen 2 — Video Reel



Technology Stack

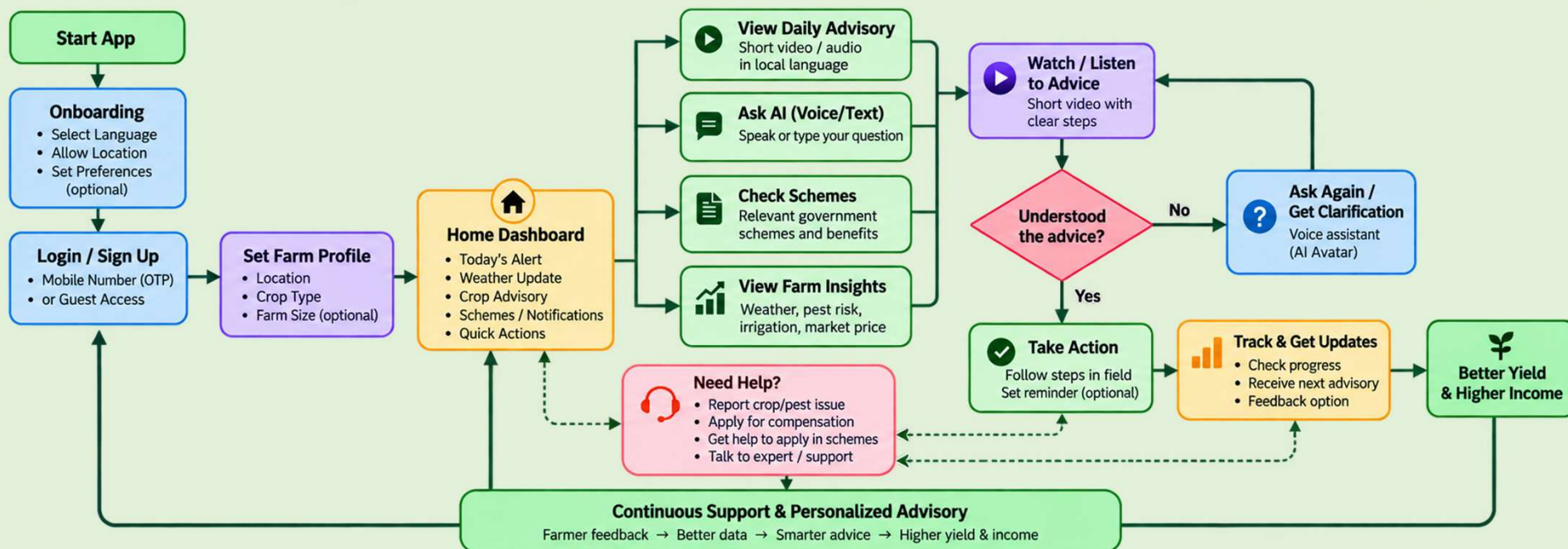


Flow Diagram

KrishiGyan – In App User Flow

From Login to Better Yield, Always With You

*Sahi Jaankari,
Behtar Kal*



Why We Need Funding — & How We Earn

Funding helps us build the technology, validate it on the ground, and reach farmers at scale.

WHERE THE MONEY IS NEEDED

AI + VIDEO ENGINE

Avatar creation, script generation, voice, lip-sync and video generation.

FIELD PILOT + VALIDATION

Local testing with farmers, crop-specific validation and feedback collection.

REACH + AWARENESS

Farmer onboarding, field demos, local campaigns and trusted creator outreach.

WEATHER + DATA + CLOUD

Reliable weather APIs, geospatial services and data-processing infrastructure.

HOW KRISHI GYAN EARNS

SMART ADVERTISING

Relevant, farmer-safe ads from verified agri brands — shown without disturbing the advisory.

FPO / COOPERATIVE PLAN

Core weather/video advice stays accessible; advanced field insights can be offered as an affordable premium.

GOVERNMENT COLLABORATION

Agriculture departments can use the platform to distribute advisories, schemes and verified public information.

LOW-COST PREMIUM

FPOs and cooperatives can pay for customized crop dashboards, alerts and group-level analytics.

THE GOAL: Keep essential farmer advice accessible • Monetize the ecosystem around it
Reinvest into better local advisory

IMPACT AND BENEFITS

Potential Impact on Target Audience



Farmers

Simple, local-language advice helps farmers understand weather and crop decisions easily.



Small & Marginal Farmers

Timely, hyper-local guidance can help reduce avoidable crop losses and input wastage.



FPOs & Cooperatives

One platform to share crop advisories, alerts and useful information with farmer groups.



Government

Better last-mile delivery of agricultural advisories, schemes and compensation information.



Benefits of the solution

Technological & Educational Benefits

Uses AI, weather data, voice and video to convert complex agricultural information into simple learning.

Environmental Benefits

Helps farmers make better decisions about irrigation, spraying and crop planning, reducing avoidable input costs.

Social Benefit

Regional-language, visual and voice-based guidance reduces the digital literacy barrier for rural farmers..

Economic benefits

Timely, actionable advice can reduce avoidable input costs, crop losses, and improve farm profitability.

12–33% Higher Profit

Indian studies found higher profits among farmers adopting weather-based agromet advisories compared with non-adopters.

Source: Journal of Agrometeorology, 2020.

KrishiGyan aims to make this proven advisory benefit more accessible through simple and visual communication.

RESEARCH AND REFERENCES



Supporting Research

1. Article: Sustainable agriculture in Northeastern India: how do tribal farmers perceive and respond to climate change?

Link : [source](#)

2. Only 30% Farmers Use Digital Advisory Satellite/mobile-based farming advisory tools

Link : [source](#)

3. 86.67% Aware, But Reach Still Uneven
A Karnataka study found 86.67% of farmers were aware of climate change, and advisory adoption was high among those reached — but access to advisory units remained limited overall.

Link: [source](#)



Market Research

1. A 2025 Bihar study estimated benefits from adopting weather-based advisories ranging from ₹1,190–₹5,760/ha for cereals and much higher amounts for some vegetable/fruit crops.

Link: [source](#)

2. AgroStar says its platform has handled 15 million agronomy queries and 400 million agronomy-video views, across 11 local languages.

Link: [source](#)

3. DeHaat reported roughly ₹3,000 crore operating revenue in FY25 and serves around 13 million farmers; its model combines inputs, market access and agricultural advisory.

Link : [source](#)



Technological Research

1. [WhatsApp Cloud API — Official Meta Documentation](#)

2. [Qdrant Vector Database — Official Qdrant Documentation](#)

3. [BHASHINI — Official BHASHINI Developer Documentation](#)

4. [IMD API — Official IMD API Reference](#)

5. [ElevenLabs — Official Documentation](#)

6. [SadTalker — Official GitHub / Documentation](#)